



# AI Engineer — Generative AI & Machine Learning

**TEIXEIRA Alexandre**

AI Engineer, University of Technology of Compiègne (UTC)

• [alexandretei13@gmail.com](mailto:alexandretei13@gmail.com) • [Teixeira · GitHub](#) • (+33)7.86.01.55.04 • [Portfolio](#)

## EDUCATION

**Master's-level Engineering Degree in Computer Science University of Technology of Compiègne (UTC),** 2026

- Relevant coursework: Linear and Nonlinear Optimization, Operations Research, Artificial Intelligence, Logic and Search-Based Problem Solving, Linear Programming, Cybersecurity.

**Exchange Program in Artificial Intelligence Shanghai University — UTSEUS China** Sep. 2024 – Jan. 2025

- Relevant coursework: Natural Language Processing, Generative AI, Data Analysis, Data Visualization.

## PROFESSIONAL EXPERIENCE

**AI Platform for Meeting Summarization and Audio Generation – SNCF** Sept. 2025 - Feb. 2026

- Developed the solution using **React, Azure AI Services, Docker, and backend services.**
- Designed and integrated an AI audio processing pipeline combining **Speech-to-Text, LLM-based summarization, and Text-to-Speech** to transform meetings into structured, actionable content.
- Industrialized the deployment workflow using **Jenkins, Harbor, and Artifactory.**

**AI Tool for Project Risk Analysis – Renault** Mars.2025 - Juin 2025

- Designed an AI-powered tool to support project risk analysis using LLMs, RAG pipelines, and vector databases.

**Generative AI Solution for DevOps Ticketing Support – SNCF** Sep. 2023 - Fév. 2024

- Contributed to one of the first internal generative AI solutions at SNCF, at a time when RAG-based enterprise applications were still emerging.
- Designed and deployed a RAG-based assistant connected to internal documentation and DevOps support tickets to help resolve technical incidents.

## SELECTED PROJECTS

**Comparative Study of CNN and Transformer Architectures University College London** Feb. 2026 - Pres.

- Compared CNN and Transformer-based architectures for 3D medical image analysis.
- Trained and evaluated early prototype models to assess performance, complexity, and inference constraints.

**Game of Thrones Character Search Engine & Graph Visualization :** 2025

- Built a search engine for Game of Thrones characters using web scraping, NLP and text processing techniques.

**AlphaZero from Scratch — Gopher & Dodo Games :** 2024

- Implemented an AI agent inspired by AlphaZero for the Gopher and Dodo games, combining Monte Carlo Tree Search, self-play and deep reinforcement learning.
- Designed a matrix-based game state encoding and developed a PyTorch residual neural network with policy and value heads.

## TECHNICAL SKILLS

**AI / LLM Engineering:** PyTorch, TensorFlow, Hugging Face, vLLM, vector databases, embeddings, LLM deployment

**Data Science & Scientific Computing:** Python, Pandas, NumPy, Matplotlib, data processing, statistical analysis

**Machine Learning / Deep Learning:** CNNs, Transformers, ResNet architectures, Scikit-learn, model training and evaluation, feature engineering, data preprocessing

**Cloud / DevOps:** Azure AI Services, Azure OpenAI, Docker, Kubernetes, Jenkins, Harbor, Artifactory, GitLab CI/CD

**Web / Backend:** React, FastAPI, Flask, Pydantic, PostgreSQL, REST APIs, SQL.

## LANGUAGES

**French:** native, **English, Chinese:** Basic knowledge

## EXTRACURRICULAR EXPERIENCE

**Participant — Founder Breakfast, Shanghai** Sept.2024 - Jan.2025

- Engaged with Shanghai's entrepreneurial ecosystem through discussions with founders and exposure to emerging AI-driven innovations.

**Partnership Manager — Imaginarium Festival** Sep. 2022 - Juin.2023

Sourced and managed partnerships for a large-scale student festival with a budget of 600,000€.